

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid-rail divider).

The diagrams show three different circuit configurations:

- Diagram 1:** A 10k resistor (R9) is connected in series between the VG_DIV signal line and ground (GND).
- Diagram 2:** A 100uF/50V capacitor (C15) is connected in parallel between the VG_DIV signal line and ground (GND).
- Diagram 3:** A 10k resistor (R9) is connected in series between the VG_DIV signal line and a 100uF/50V capacitor (C15), which is then connected to ground (GND).

Figure 1: Schematic diagram of the test setup. The diagram shows a J1 connector with pins 1, 2, and GND. Pin 1 is connected to IN_L, pin 2 to GND, and GND to IN_L. A signal source is connected to IN_L and A_L. A resistor R_{b1} (100k) is connected between A_L and VGND.

Diagram illustrating the C1 select signal structure. The signal is composed of three parts: C1A_SEL, C1B_SEL, and C1C_SEL. The signal is shown as a vertical bar with a red segment at the top. The signal is labeled as C1A_SEL, C1B_SEL, and C1C_SEL. The signal is shown as a vertical bar with a red segment at the top. The signal is labeled as C1A_SEL, C1B_SEL, and C1C_SEL.

Legend for C1 select signal:

Signal	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6
JP1	1	2	3	4	5	6
C1 select	1	2	3	4	5	6

OUT1 1
 OUT2 2
 SW_COM 3

J5
 POLARITY SW

SW_COM
 C_out1
 10uF/50V
 POT_TOP

OUT_TIP 1
OUT_RING 2
OUT_GND 3

J3
OUT 3.5mm

[illegible]

D1 = power lamp on V12 (proves the LM7812, not just VIN).
D2 = INVERTED lamp, fed from SW2's unused second pole via J7.
Both ~2 mA. No LED touches a signal node.

Off-board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
 Process: single-sided PCB, bottom copper only, board outline 100 x 100 mm.
 Copper is isolation-milled on the CNC with an 0.8 mm flat end mill; the fiber
 laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
 The 0.8 mm tool width is a hard floor on every copper-to-copper gap. Fixed-pitch
 pads sit below the 0.85 mm target and carry a .xicad.dru exception: DIP-14
 0.94 mm (passes), 2.54 mm headers and the T0-220 0.84 mm (do not).
 Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c -> C1_1/C1_2/C1_3, C2a/C2b/C2c -> C2_1/C2_2/C2_3, R1a/R1b -> R1_1/R1_2, C_out -> C_out1.

Single-sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono-summing 2nd-order Sallen-Key low-pass, single +12 V rail

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Title: Bose Companion 5 sub crossover

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